

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2013-14

Zoology**Paper: ZOB-503 — Mammalian Endocrinology & Developmental Biology**

Time: Three hours

Full Marks: 70

Note: Answer three questions from Section-A and three questions from Section-B. Question No. 1 in Section-A and Question No. 5 in Section-B are compulsory. Use a separate answer book for each Section. The figures in the right-hand margin indicate marks.

SECTION – A**(Mammalian Endocrinology) (Marks: 35)**

Note: Answer three questions including Question No. 1 which is compulsory.

Question 1. Answer the following parts:

(a) Write whether the following statements are True or False. Justify your answer: $3 \times 2 = 6$

- (i) Hypophysial portal vessel is a vascular link between median eminence and pars distalis.
- (ii) Vaginal smear shows mainly leucocytes during oestrus phase of the oestrous cycle.
- (iii) Diabetes insipidus is caused due to deficiency of relaxin.

(b) Write the best option from the multiple choices given against each question: $4 \times \frac{1}{2} = 2$

(i) Gastrin is secreted from:

- (1) Antral mucosa
- (2) Jejunum
- (3) Pancreas
- (4) Gall bladder

(ii) Melatonin is synthesized from:

- (1) L-Tyrosine
- (2) L-Tryptophan
- (3) Melanin
- (4) Histamine

(iii) α -adrenergic receptor stimulation causes:

- (1) Smooth muscle relaxation
- (2) Smooth muscle contraction

- (3) Both contraction and relaxation of smooth muscles
- (4) No effect on smooth muscle
- (iv) In circulation thyroxine is bound maximally by:
 - (1) Thyroxine-binding globulin
 - (2) Transthyretin
 - (3) Albumin
 - (4) Not bound to any proteins
- (c) Define the following: [3 × 1 = 3]
 - (i) Second messenger
 - (ii) Iodide pump
 - (iii) Addison's disease.

Question 2. Describe the biological actions of Testosterone and regulation of its secretion in the testis. [12]

Question 3. Explain the regulation of glucocorticoid secretion. Why is the glucocorticoid essential for adrenaline synthesis? Comment upon General Adaptation Syndrome. [6 + 2 + 4 = 12]

Question 4. Write short notes on the following: [3 × 4 = 12]

- (a) Goitre
- (b) Diabetes mellitus
- (c) Oxytocin.

SECTION – B
(Developmental Biology) (Marks: 35)

Note: Answer three questions including Question No. 5 which is compulsory.

Question 5. Answer the following parts:

(a) State whether the following statements are True or False, giving brief reasons within 3–4 lines: [2 × 3 = 6]

- (i) The primitive streak is derived from the anterior epiblast cells and central cells of the posterior marginal zone.
- (ii) The cells of outer layer of optic cup proliferates and generates light sensitive photo receptor neurons.
- (iii) Magnesium and potassium act as the initiator of the cortical granule reaction.

(b) Select the best options from the following: [6 × $\frac{1}{2}$ = 3]

- (i) In sea urchin, cleavage is:
 - (1) Spiral
 - (2) Radial
 - (3) Bilateral
 - (4) Rotational
- (ii) β -catenin is a key player in the formation of:
 - (1) Ventral tissues
 - (2) Anterior tissues
 - (3) Dorsal tissues
 - (4) Posterior tissues
- (iii) Pax 6 gene expression is seen in the:
 - (1) Ectoderm
 - (2) Mesoderm
 - (3) Head ectoderm
 - (4) Head mesoderm
- (iv) Optic vesicle is acting as an:
 - (1) Inducer
 - (2) Responder
 - (3) Inhibitor
 - (4) All of the above

(v) Acrosomal process is formed during:

- (1) Mouse fertilization
 - (2) Sea urchin fertilization
 - (3) Both (i) and (ii)
 - (4) None of the above
- (vi) Conversion of T_4 to T_3 is by:
- (1) Deiodinase II
 - (2) Deiodinase III
 - (3) Deiodinase IV
 - (4) All of the above

(c) Define the following:

[2 × 1 = 2]

- (i) Bindin
- (ii) Induction.

Question 6. Answer the following parts:

[6 + 6 = 12]

- (a) Discuss morphogenetic gradients in sea urchin eggs.
- (b) Describe the mechanism of Hormonal control of amphibian metamorphosis.

Question 7. Write short notes on the following:

[3 × 4 = 12]

- (a) Gastrulation in sea urchin
- (b) Embryonic stem cells
- (c) Fate map of frog.

Question 8. Describe the mechanism of induction and determination during vertebrate eye formation.

[12]

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